Cardiovascular System (CARD) · NPLEX-P1-CARD-17

CV-17 Cardiovascular Pathogens

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-17-01-C01-BD1

Brain dump 1 · foundational retrieval: Rebuild the four endocarditis cocci

A competent response contains

· Staphylococcus aureus: grapelike clusters because it divides in several planes, catalase positive, coagulase positive, beta-hemolytic.

· S. aureus virulence factors: bound coagulase (clumping factor) binds fibrinogen, free coagulase activates prothrombin so fibrin coats the organism, fibronectin-binding proteins allow attachment to and entry into endothelial cells, protein A binds the Fc region of IgG, and pore-forming alpha toxin with tissue-degrading enzymes destroys leaflets within days.

· Viridans streptococci: chains because they divide in one plane, catalase negative, alpha-hemolytic from hydrogen peroxide oxidizing hemoglobin, optochin resistant and bile insoluble, which separates them from Streptococcus pneumoniae.

· Viridans virulence factor: dextran, a glucan polymer made from dietary sucrose, binds the platelet-fibrin deposit on an already damaged valve, and the organism becomes buried in it and shielded from neutrophils.

· Enterococcus faecalis: pairs and short chains, catalase negative, Lancefield group D antigen, grows in 6.5% NaCl and in bile, hydrolyzes esculin; surface adhesins and biofilm plus intrinsic resistance to many antibiotics.

· Staphylococcus epidermidis: clusters, catalase positive, coagulase negative, novobiocin sensitive, low virulence, and its one important factor is a polysaccharide adhesin that builds biofilm excluding neutrophils and antibodies.

· S. aureus acts on normal valves and destroys them acutely; viridans streptococci and enterococci act on damaged valves over weeks to months; S. epidermidis acts on prosthetic valves, pacemaker leads, and catheters indolently.

· Arrangement (clusters against chains) and catalase separate staphylococci from streptococci and enterococci, and coagulase separates S. aureus from S. epidermidis.

Distinctions that matter

· Catalase is a staphylococcal property; streptococci and enterococci are catalase negative, so catalase comes before coagulase in the reasoning.

· Adhesion mechanism decides which valve an organism can take: fibrinogen and fibronectin binding works on near-normal endothelium, dextran binding needs a platelet-fibrin deposit, and biofilm needs a foreign surface.

Common misconceptions

· Treating alpha hemolysis as partial lysis by a toxin. It is a greenish discoloration produced when the organism's hydrogen peroxide oxidizes hemoglobin.

· Calling S. epidermidis harmless because it is a skin commensal. On a prosthetic valve its biofilm makes it the classic cause of early prosthetic valve endocarditis, although a single positive culture is often contamination.

Where it is taught: Section C01, organism subsections and the table of gram-positive cocci of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C01-BD2

Brain dump 2 · integration and application: Two patients, two valves, two time courses

A competent response contains

· Patient one: Staphylococcus aureus from skin flora entering the venous blood, so the tricuspid valve is reached first; clumping factor and fibronectin-binding proteins attach it to minimally injured or normal endothelium, and alpha toxin with tissue-degrading enzymes perforates the leaflet within days.

· Patient two: viridans streptococci from mouth flora entering during the brief bacteremia of dental work; turbulent flow across the bicuspid valve had already injured the endothelium and laid down a sterile platelet-fibrin deposit, and dextran binds the organism into it.

· Viridans streptococci invade poorly and are weakly toxic, so they need an abnormal valve and give subacute disease over weeks to months.

· Protein A binds the Fc region of IgG and holds the antibody in an orientation that cannot opsonize or fix complement; bound coagulase and free coagulase coat the organism in fibrin, which hinders phagocytosis.

· A sterile platelet-fibrin deposit gives a poorly invasive organism a surface it can stick to and then be buried in, out of reach of neutrophils.

· Patient one is the one expected to have an abscess and perforation, because S. aureus destroys tissue directly with toxins and enzymes.

Distinctions that matter

· Acute against subacute follows from the organism's toxins, not from how long the patient waited to be seen.

· Right-sided disease in injection drug use follows the route of entry: skin organisms reach the right heart first through venous blood.

Common misconceptions

· Expecting a damaged valve in every case of endocarditis. S. aureus can colonize a normal valve.

· Assuming fibrin around the organism helps the host. Coagulase-generated fibrin shields the bacterium from phagocytes.

Where it is taught: Section C01, S. aureus and viridans streptococci subsections and the stop and predict box of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C01-CP01

Comparison table: The four gram-positive cocci of infective endocarditis

A competent response contains

· Arrangement: S. aureus and S. epidermidis in clusters; viridans streptococci in chains; E. faecalis in pairs and short chains.

· Catalase: positive for both staphylococci, negative for viridans streptococci and E. faecalis.

· Other bench tests: S. aureus coagulase positive and beta-hemolytic; viridans streptococci alpha-hemolytic, optochin resistant, bile insoluble; E. faecalis group D antigen, grows in 6.5% NaCl and bile, hydrolyzes esculin; S. epidermidis coagulase negative and novobiocin sensitive.

· Virulence factors: clumping factor, fibronectin-binding proteins, protein A, alpha toxin and tissue-degrading enzymes for S. aureus; dextran for viridans streptococci; adhesins, biofilm, and intrinsic drug resistance for E. faecalis; polysaccharide adhesin and biofilm for S. epidermidis.

· Attachment: S. aureus binds fibrinogen and fibronectin on minimally injured endothelium; viridans streptococci bind an existing platelet-fibrin deposit; E. faecalis uses surface adhesins and biofilm; S. epidermidis builds biofilm on a foreign surface.

· Valve required: normal or minimally injured for S. aureus; scarred or congenitally abnormal for viridans streptococci; usually abnormal for E. faecalis; prosthetic valve or device for S. epidermidis.

· Portal of entry: skin, including injection drug use, for S. aureus with the tricuspid valve reached first; mouth with dental bacteremia for viridans streptococci; urinary or gastrointestinal tract, often after instrumentation in older adults, for E. faecalis; skin at the time of surgery or device insertion for S. epidermidis.

· Course: acute with abscess and perforation for S. aureus; subacute over weeks to months for viridans streptococci and E. faecalis; indolent, usually within the first year after valve surgery, for S. epidermidis.

Distinctions that matter

· Both staphylococci are catalase positive, so coagulase is the test that tells them apart.

· E. faecalis and viridans streptococci are both catalase negative, and growth in 6.5% NaCl and bile with esculin hydrolysis marks the enterococcus.

Common misconceptions

· Filling the valve row with abnormal for every organism. The S. aureus row is the exception that explains acute endocarditis on a normal valve.

· Listing biofilm for every organism. Biofilm is the defining factor for S. epidermidis on devices and contributes for E. faecalis, but it is not what viridans streptococci or S. aureus rely on.

Where it is taught: Section C01, table of gram-positive cocci of infective endocarditis of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C02-BD1

Brain dump 1 · foundational retrieval: Why blood cultures stay negative

A competent response contains

· Culture-negative endocarditis is endocarditis by clinical and echocardiographic findings with no growth in routine blood cultures.

· The commonest cause is antibiotics given before the cultures were drawn, which suppresses growth of any organism.

· HACEK: Haemophilus species, mainly H. parainfluenzae; Aggregatibacter; Cardiobacterium hominis; Eikenella corrodens; Kingella kingae. They are fastidious gram-negative rods and coccobacilli of the normal oropharyngeal flora, need enriched media and added CO2, grow slowly, and tend to form large friable vegetations that embolize. Modern automated systems usually recover them within about 5 days.

· Coxiella burnetii is a small gram-negative obligate intracellular bacterium that multiplies inside the acidic phagolysosome of macrophages; a spore-like small-cell form survives drying, so infection follows inhaling dust or aerosols contaminated with the placenta and amniotic fluid of cattle, sheep, and goats, with no vector involved.

· Coxiella does not grow in routine culture, so diagnosis rests on serology, with high titers against phase I antigen marking chronic infection, and chronic endocarditis occurs with damaged or prosthetic valves, vascular grafts, or weak immunity, often with small or absent vegetations.

· Bartonella species are fastidious, facultatively intracellular gram-negative rods. B. quintana is carried by the human body louse, and B. henselae comes from cats through scratches or cat fleas. Diagnosis uses serology, PCR, or silver-stained valve tissue.

· Streptococcus gallolyticus is a group D streptococcus of the colon that is bile-esculin positive but does not grow in 6.5% NaCl, which is what separates it from Enterococcus.

Distinctions that matter

· Fastidious means slow growth on enriched media; obligate intracellular means no growth on cell-free routine media at all, which is why HACEK eventually grows and Coxiella does not.

· Coxiella needs no vector, while Bartonella quintana needs the body louse, so the exposure history separates them.

Common misconceptions

· Reading culture-negative as unusual organism first. Prior antibiotics are the commonest explanation.

· Expecting large vegetations in Q fever endocarditis. Vegetations are often small or absent, which is one reason the diagnosis is missed.

Where it is taught: Section C02, opening paragraph, HACEK and Coxiella subsections, and the table of causes of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C02-BD2

Brain dump 2 · integration and application: Match the exposure to the organism and read the bench results

A competent response contains

· Goat farm with kidding: Coxiella burnetii, inhaled from aerosols of birth products; it is an obligate intracellular organism that multiplies in the macrophage phagolysosome and does not grow on routine media, so serology is used, with phase I titers marking chronic infection.

· Living outdoors with body lice: Bartonella quintana, a fastidious gram-negative rod that grows very slowly, identified by serology, PCR, or silver stain of valve tissue.

· Poor dental health with large friable embolizing vegetations: a HACEK organism from oropharyngeal flora, fastidious and slow growing and needing enriched media with added CO2.

· Recent antibiotic: any organism with its growth suppressed, which is the commonest cause of culture-negative endocarditis.

· The second patient's isolate is Streptococcus gallolyticus: chains with a negative catalase place it among streptococci and enterococci, bile esculin hydrolysis marks a group D organism, and failure to grow in 6.5% NaCl excludes Enterococcus.

· Its source is the colon, and a colorectal adenoma or carcinoma disrupting the mucosal barrier is what lets it enter the blood, so its isolation calls for evaluation of the colon even without bowel symptoms.

· Had the isolate grown in 6.5% NaCl it would be Enterococcus, with a urinary or gastrointestinal source, often after instrumentation in an older adult.

Distinctions that matter

· Bile esculin positivity is shared by Enterococcus and S. gallolyticus; the salt tolerance test is the one that separates them.

· Exposure history does most of the work in culture-negative endocarditis, because each organism has a characteristic source: livestock aerosols, body lice, cats, or the mouth.

Common misconceptions

· Treating S. gallolyticus bacteremia as a finding about the heart alone. It is also a finding about the colon.

· Expecting Coxiella to come from an animal bite or a tick. Transmission is by inhalation, with no vector.

Where it is taught: Section C02, table of causes, the S. gallolyticus subsection, and the worked example of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C02-Q01

Board-style question: A group D organism from the bench tests

Correct answer: C. Streptococcus gallolyticus

Why it is right: Streptococcus gallolyticus is a group D streptococcus of the colon that hydrolyzes esculin in bile but, unlike Enterococcus, does not grow in 6.5% NaCl, and it reaches the blood through a colorectal adenoma or carcinoma that has broken the mucosal barrier.

A is wrong: Enterococcus faecalis is also bile-esculin positive but grows in 6.5% NaCl and in bile, which this isolate did not.

B is wrong: Streptococcus mitis is a viridans streptococcus, so it is alpha-hemolytic rather than nonhemolytic and is not bile-esculin positive; it follows dental bacteremia on a damaged valve.

D is wrong: Staphylococcus epidermidis grows in clusters and is catalase positive, and it infects prosthetic valves and devices by biofilm rather than native valves in this pattern.

A competent response contains

· Chains with a negative catalase place the isolate among the streptococci and enterococci.

· Esculin hydrolysis in bile marks a group D organism, which leaves Enterococcus and Streptococcus gallolyticus.

· Failure to grow in 6.5% NaCl excludes Enterococcus, leaving S. gallolyticus, whose source is the colon.

Distinctions that matter

· Both group D organisms are bile-esculin positive, and only Enterococcus grows in 6.5% NaCl.

Common misconceptions

· Stopping at group D and naming Enterococcus, which is the commoner group D isolate but does not fit the salt result.

Where it is taught: Section C02 of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C03-BD1

Brain dump 1 · foundational retrieval: The two human forms and every route of transmission

A competent response contains

· Trypanosoma cruzi is a flagellated protozoan and the cause of Chagas disease, also called American trypanosomiasis.

· Trypomastigote: extracellular, flagellated, circulates in blood, invades cells, and does not divide in humans.

· Amastigote: intracellular, rounded, no external flagellum, divides by binary fission in the cytoplasm after the parasite escapes its vacuole.

· Multiplication in the human happens only inside host cells, and the favored cells are macrophages, cardiac myocytes, smooth muscle, and neurons of the enteric plexuses.

· The vector is the triatomine bug, also called the reduviid or kissing bug, which lives in the cracks of poorly built houses in rural Mexico, Central America, and South America and feeds at night, often on the face.

· The infective metacyclic trypomastigotes are in the bug's feces, not its saliva, and the bug defecates during or soon after the blood meal, so the feces must be rubbed into the bite wound, conjunctiva, or mucosa; the bite alone does not transmit.

· Other routes: blood transfusion, organ transplant, across the placenta, food or juice contaminated with bugs or their feces, and laboratory accidents. Oral outbreaks deliver a large inoculum and tend to cause severe acute myocarditis.

· The acute phase lasts several weeks to about 2 months, is usually mild, and shows many trypomastigotes on a stained blood smear.

· Entry through the conjunctiva gives painless swelling of one eyelid, the Romana sign; entry through skin gives a local inflammatory nodule, a chagoma.

Distinctions that matter

· T. cruzi trypomastigotes circulate but do not divide, unlike the African trypanosomes, which multiply in the blood.

· The parasite is in the feces rather than the saliva, which is why scratching the bite matters more than the bite.

Common misconceptions

· Saying the bug injects the parasite as it bites. The feces deposited beside the bite are rubbed in afterwards.

· Describing the amastigote as flagellated. It is rounded with no external flagellum, and it is the only dividing form in the human.

Where it is taught: Section C03, opening paragraph, the life cycle, and the paragraph on the two human forms of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C03-BD2

Brain dump 2 · integration and application: Predict chronic Chagas disease from the cells the parasite lives in

A competent response contains

· She has been in the indeterminate phase, which is lifelong in most people, with few circulating parasites, so diagnosis depends on serology rather than a blood smear.

· About 20 to 30% of infected people develop chronic disease over decades.

· Persisting amastigotes and the inflammation they provoke destroy myocytes, which are replaced by fibrosis.

· Cardiac result: dilated cardiomyopathy with reduced contractility and enlarged chambers, a thin apical aneurysm that can hold thrombus, and conduction disease, typically right bundle branch block with left anterior fascicular block.

· Loss of myenteric neurons leaves the esophagus and colon unable to relax and propel their contents, so they dilate into megaesophagus and megacolon.

· The organs dilate because the neurons that coordinate relaxation and propulsion are gone, so contents accumulate behind a segment that will not relax and the wall stretches over time.

· Trypomastigotes circulate and amastigotes persist in tissue for life, so transfusion, transplant, and placental transmission occur far from any vector, which is why blood banks and transplant programs screen for infection.

Distinctions that matter

· Cell tropism predicts the lesion: cardiac myocytes give cardiomyopathy and conduction disease, enteric neurons give megaesophagus and megacolon.

· The acute phase is diagnosed by seeing the parasite; the indeterminate and chronic phases are diagnosed by serology.

Common misconceptions

· Expecting chronic disease in everyone infected. Most people stay in the indeterminate phase.

· Attributing the chronic lesions to the parasite alone. Myocyte destruction and the T-cell response together drive the fibrosis.

Where it is taught: Section C03, paragraphs on transmission without the bug and on the phases of disease of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C03-RT01

Retrieval exercise: Rebuild the Trypanosoma cruzi life cycle

A competent response contains

· Step 1: the bug ingests blood trypomastigotes during a blood meal.

· Step 2: in the bug midgut, epimastigotes multiply.

· Step 3: in the hindgut, metacyclic trypomastigotes form and pass in the feces.

· Step 4: the feces are rubbed into the bite wound, the conjunctiva, or a mucosal surface.

· Step 5: trypomastigotes invade host cells and become amastigotes.

· Step 6: amastigotes divide by binary fission in the cytoplasm.

· Step 7: amastigotes become trypomastigotes and the cell ruptures.

· Step 8: blood trypomastigotes infect new cells or a feeding bug.

· Steps 1 to 3 happen in the insect and steps 4 to 8 in the human.

· The two multiplying steps are the epimastigote in the bug midgut and the amastigote inside the human cell.

· Parasite numbers in blood rise because each invaded cell releases many trypomastigotes when it ruptures, even though the circulating form itself never divides.

Distinctions that matter

· Three forms, three places: epimastigote in the insect gut, amastigote inside human cells, trypomastigote in blood and in insect feces.

· Only the metacyclic trypomastigote in the hindgut feces is infective to humans.

Common misconceptions

· Putting the dividing form in the bloodstream. Division in the human happens only intracellularly.

· Leaving out the rubbing-in step, which is where transmission actually occurs.

Where it is taught: Section C03, life cycle diagram of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C04-BD1

Brain dump 1 · foundational retrieval: Two tick-borne bacteria from the ground up

A competent response contains

· Borrelia burgdorferi is a spirochete: a long, thin, helical bacterium whose flagella lie in the periplasm between the inner and outer membranes, and their rotation turns the whole cell like a corkscrew so it can move through connective tissue.

· Its outer membrane lacks classic lipopolysaccharide and is rich in lipoproteins, and its genome is a linear chromosome plus many linear and circular plasmids that carry most of the surface protein genes.

· Rickettsia rickettsii is a small gram-negative coccobacillus and is obligate intracellular because it imports ATP and other metabolites from host cytoplasm and cannot make enough of its own coenzyme A and NAD+.

· Stains: Borrelia with Giemsa, Wright, or silver; Rickettsia with Giemsa or Gimenez. Borrelia is too thin to see well on Gram stain and Rickettsia stains poorly with it.

· Borrelia vectors are Ixodes ticks, Ixodes scapularis in the eastern United States and Ixodes pacificus in the West; Rickettsia rickettsii is carried by Dermacentor ticks, the American dog tick and the Rocky Mountain wood tick.

· Reservoirs: the white-footed mouse for Borrelia, with larval ticks acquiring the organism from it and deer feeding adult ticks without being a significant reservoir; the tick itself for Rickettsia rickettsii, which passes it to its offspring through the eggs.

· Nymphal Ixodes ticks transmit most human Lyme disease.

· Despite the name, most United States cases of Rocky Mountain spotted fever occur in southeastern and south-central states.

Distinctions that matter

· Borrelia is extracellular and invades tissue mechanically; Rickettsia rickettsii is obligate intracellular in endothelium.

· Borrelia needs a mammalian reservoir, while the Dermacentor tick is both vector and reservoir for Rickettsia rickettsii.

Common misconceptions

· Placing spirochete flagella on the outside of the cell. They are periplasmic, which is what makes the whole cell rotate.

· Naming deer as the Lyme reservoir. Deer feed adult ticks; the white-footed mouse is the reservoir.

Where it is taught: Section C04, Borrelia and Rickettsia subsections and the comparison table of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C04-BD2

Brain dump 2 · integration and application: Explain the block and explain the rash

A competent response contains

· Patient one: Borrelia burgdorferi from an Ixodes tick, with the expanding lesion of erythema migrans; spread through blood carries the spirochete to the heart, where spirochetes with a lymphocyte and plasma cell infiltrate involve the myocardium and conduction tissue.

· Lyme carditis most often produces atrioventricular block at the level of the AV node, it can change degree within hours as the inflammatory infiltrate shifts, and it usually resolves as inflammation subsides.

· Patient two: Rickettsia rickettsii from a Dermacentor tick, infecting the endothelium of small vessels of the skin, brain, lungs, and kidneys.

· OmpA and OmpB bind endothelial cells and trigger uptake, the bacterium lyses the phagosome and multiplies in the cytoplasm, then polymerizes host actin into a tail that pushes it into the next endothelial cell, so infection travels along small vessels without re-entering the blood.

· Injured endothelium becomes leaky and procoagulant, which gives edema, loss of intravascular volume with hypotension, and platelet consumption; the rash begins as macules on the wrists and ankles a few days after fever, spreads to the trunk and onto the palms and soles, and becomes petechial as vessel damage progresses.

· Early tick removal works because Borrelia in an unfed tick gut expresses OspA, which anchors it to the gut, and it must switch to OspC, multiply, and migrate to the salivary glands before it can be injected, which usually takes 36 to 48 hours of attachment.

Distinctions that matter

· Borrelia damages conduction tissue by direct invasion with an inflammatory infiltrate; Rickettsia damages the vessel wall from inside endothelial cells.

· Lyme block is at the node and usually reversible, which is why its degree can vary within hours.

Common misconceptions

· Thinking the spotted fever rash begins centrally. It starts peripherally on the wrists and ankles and moves inward, reaching palms and soles.

· Assuming any tick bite transmits Borrelia immediately. The surface switch and migration take time, so a tick found and removed on the first day rarely transmits.

Where it is taught: Section C04, Borrelia flow and host paragraph, Rickettsia paragraphs, and the stop and predict box of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C04-RT01

Retrieval exercise: Rebuild the Borrelia surface switch from tick to host

A competent response contains

· Step 1: in the unfed tick gut, Borrelia expresses OspA, which anchors it to the gut.

· Step 2: the blood meal warms the tick and supplies nutrients.

· Step 3: OspA falls, OspC rises, and the spirochetes multiply.

· Step 4: spirochetes migrate to the tick salivary glands.

· Step 5: OspC binds the tick salivary protein Salp15, which shields the spirochete.

· Step 6: injection into the skin, usually after 36 to 48 hours of attachment.

· Step 7: in the host, VlsE variation keeps specific antibody behind.

· The trigger for the switch is the blood meal itself, through warming and nutrients.

· In the host, recombination moves segments of silent vls cassettes into the expressed vlsE gene, so the surface lipoprotein VlsE keeps changing and antibody raised against the earlier version no longer binds.

Distinctions that matter

· OspA is for life in the tick gut and OspC is for transmission and early infection, so the two proteins mark two different environments.

· Antigenic variation through VlsE is a host-phase strategy and has nothing to do with the tick.

Common misconceptions

· Treating the delay in transmission as travel time through the skin. The delay is the surface switch, replication, and migration to the salivary glands inside the tick.

· Describing VlsE change as mutation. It is recombination from silent cassettes into the expressed gene.

Where it is taught: Section C04, Borrelia surface switching flow and the paragraph after it of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C05-BD1

Brain dump 1 · foundational retrieval: Flaviviruses and filoviruses side by side

A competent response contains

· Flaviviruses: yellow fever virus and dengue virus, small spherical enveloped virions with a positive-sense single-stranded RNA genome translated as one polyprotein that proteases cut.

· Filoviruses: Ebola virus and Marburg virus, long filamentous enveloped virions with a negative-sense single-stranded RNA genome, so the virion must carry an RNA-dependent RNA polymerase.

· A positive-sense genome is itself mRNA, so naked genome RNA is infectious; a negative-sense genome is complementary to mRNA and naked genome RNA is not infectious.

· Reservoirs: humans for dengue and monkeys in the forest cycle for yellow fever; fruit bats for filoviruses, proven for Marburg virus with Rousettus bats and strongly suspected for Ebola virus.

· Transmission: Aedes mosquito bite for the flaviviruses, with Aedes aegypti a day-biting urban mosquito; contact with blood and body fluids for filoviruses, with no vector.

· First cells: monocytes and dendritic cells for the flaviviruses, with hepatocytes for yellow fever; macrophages, monocytes, and dendritic cells for filoviruses, with endothelium and hepatocytes later.

· Yellow fever: hepatocytes in the midzone of the lobule, zone 2, die by apoptosis and appear as eosinophilic Councilman bodies; lost hepatocyte function gives jaundice and reduced clotting factor synthesis, which with endothelial damage produces bleeding, including black vomit, and renal failure.

· Dengue: four serotypes DENV-1 to DENV-4 infecting dendritic cells, monocytes, and macrophages; severe dengue is a plasma leak syndrome that peaks as fever falls, with hemoconcentration, low platelets, and shock, and the secreted protein NS1 adds to leak by damaging the endothelial glycocalyx.

Distinctions that matter

· Both families end in endothelial leak and coagulopathy, but by different routes: plasma leak and antibody-assisted entry in dengue, hepatocyte apoptosis in yellow fever, and interferon blockade with cytokine storm in the filoviruses.

· Strand sense predicts what the virion must bring with it, which is why only the negative-sense virion carries a polymerase.

Common misconceptions

· Calling filoviruses arthropod-borne. They spread by direct contact with blood and body fluids, including those of the dead, and can persist for months in semen.

· Placing yellow fever necrosis at the center or edge of the lobule. It is the midzone, zone 2.

Where it is taught: Section C05, opening paragraph, the comparison table, and the yellow fever and dengue subsection of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C05-BD2

Brain dump 2 · integration and application: Why the second dengue infection is the dangerous one

A competent response contains

· Infection with one serotype gives lasting immunity to that serotype but only brief cross-protection against the other three.

· Years later the old antibody still binds the new serotype but cannot neutralize it, and the coated virus attaches to Fc gamma receptors on monocytes and macrophages.

· That route takes the virus in more efficiently than the virus could enter alone, so viral load rises, and entry by that route also dampens interferon responses.

· Cross-reactive memory T cells add a burst of cytokines that worsens the leak.

· An infant with waning maternal antibody has the same situation at a first infection: nonneutralizing antibody is present without protective immunity.

· A rising hematocrit with no external fluid loss means plasma has leaked out of the vessels, leaving the red cells concentrated; falling platelets add bleeding risk to shock.

· The patient gets worse as the fever breaks because that is when the critical phase of plasma leak begins.

· Ebola differs: the glycoprotein GP mediates entry into macrophages, monocytes, and dendritic cells, VP35 blocks type I interferon induction and VP24 blocks interferon signaling, infected dendritic cells fail to activate T cells and bystander lymphocytes die by apoptosis, and infected macrophages release TNF, IL-6, and tissue factor, with tissue factor triggering disseminated intravascular coagulation.

Distinctions that matter

· In dengue the antibody is the problem because it helps the virus in; in Ebola the problem is that interferon is blocked before antibody is ever made.

· Hemoconcentration reports where the plasma went, while thrombocytopenia reports the bleeding risk; they are separate findings.

Common misconceptions

· Expecting the sickest point of dengue to be the fever peak. Severe disease begins as the fever falls.

· Treating any antibody as protective. Nonneutralizing antibody plus Fc gamma receptors is a route of entry.

Where it is taught: Section C05, antibody-dependent enhancement subsection, the worked example on dengue labs, and the Ebola and Marburg subsection of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C05-PR01

Prediction exercise: Predict the course of a second dengue infection

A competent response contains

· The DENV-1 antibody binds DENV-2 but does not neutralize it, because cross-protection between serotypes is brief.

· Virus entry into monocytes and macrophages rises, because the coated virus attaches to Fc gamma receptors, which take it in more efficiently than the virus could enter alone.

· The type I interferon response in those cells falls, because entry by the Fc receptor route dampens interferon signaling, so replication is less restrained.

· Fluid leaves the vessels and the hematocrit rises, because NS1 damages the endothelial glycocalyx and cytokines from cross-reactive memory T cells increase permeability.

· The platelet count falls, which is part of the severe dengue picture of hemoconcentration, low platelets, and shock.

· Blood pressure falls and she becomes lightheaded, because the critical phase of plasma leak begins as the fever falls and intravascular volume drops.

Distinctions that matter

· Rising hematocrit with falling platelets is the signature pair: one marks plasma leak, the other marks bleeding risk.

· Entry by Fc receptor does two things at once, raising viral load and lowering interferon, so both belong in the prediction.

Common misconceptions

· Predicting improvement when the fever falls. The critical phase begins at that point.

· Explaining the raised hematocrit as dehydration from fever. Plasma has leaked into the tissues rather than been lost from the body.

Where it is taught: Section C05, antibody-dependent enhancement subsection and the worked example on dengue labs of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C06-BD1

Brain dump 1 · foundational retrieval: Coxsackie B from gut to dilated heart

A competent response contains

· Myocarditis is inflammation of the myocardium with myocyte injury, and viruses are its most commonly identified cause in high-income countries.

· Coxsackie B virus, serotypes B1 to B6, is an enterovirus in the picornavirus family: small, nonenveloped, icosahedral, with a positive-sense single-stranded RNA genome.

· Having no envelope lets it resist stomach acid, bile, and drying, so it spreads mainly by the fecal-oral route, and also through respiratory secretions, mostly in summer and fall.

· Step 1: fecal-oral or respiratory entry with replication in pharyngeal and intestinal lymphoid tissue. Step 2: viremia carries virus to the heart. Step 3: the virus binds DAF (CD55), then CAR at the intercalated discs. Step 4: the genome is translated into one polyprotein. Step 5: protease 2A cleaves eIF4G and dystrophin. Step 6: the sarcolemma tears, myocytes die and release troponin. Step 7: T-cell inflammation and fibrosis, with possible dilated cardiomyopathy.

· Entry depends on the coxsackie-adenovirus receptor (CAR), a transmembrane protein of the immunoglobulin superfamily in tight junctions and, in the heart, at the intercalated discs; many strains first bind decay-accelerating factor (DAF, CD55), which helps deliver them to CAR.

· CAR is abundant in the developing heart and is expressed again in injured myocardium, which is one reason neonatal coxsackie myocarditis is severe.

· Protease 2A cleaves eIF4G, which shuts off cap-dependent host translation, and cleaves dystrophin, which links sarcomeric actin to the dystrophin-glycoprotein complex in the sarcolemma, so the membrane tears under contractile stress.

· The virus also causes pleurodynia and aseptic meningitis, and it often inflames the pericardium as well, giving myopericarditis with pleuritic chest pain and a friction rub.

Distinctions that matter

· Translation of the genome is cap-independent through an internal ribosome entry site, which is why cleaving eIF4G stops host protein synthesis without stopping viral synthesis.

· DAF is a helper that delivers the virus; CAR is the receptor that gets it in.

Common misconceptions

· Expecting an enveloped virus because it reaches the heart through the blood. It is naked, and that is exactly why it survives the gut.

· Attributing all the damage to the immune response. Direct viral injury dominates the first days, and immune injury follows over weeks.

Where it is taught: Section C06, opening paragraph, the gut to dilated heart flow, and the paragraph on entry and protease 2A of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C06-BD2

Brain dump 2 · integration and application: Explain the findings and judge a biopsy result

A competent response contains

· Troponin appears in the blood because myocytes die when the sarcolemma tears and release their contents.

· The pain is pleuritic with a rub because coxsackie B often inflames the pericardium alongside the myocardium, which is myopericarditis.

· The rhythm is unstable because inflamed myocardium is electrically unstable, giving arrhythmias with ST and T wave changes.

· Contractility falls because myocytes have been lost.

· The protein is dystrophin, which links sarcomeric actin to the dystrophin-glycoprotein complex in the sarcolemma; the inherited disease with the same structural failure is Duchenne muscular dystrophy.

· The two differ in how the link is lost: viral protease 2A cuts the existing dystrophin, while in Duchenne muscular dystrophy the protein is not made properly in the first place, and in both cases the membrane tears under contractile stress.

· Over the following weeks the T-cell response clears virus but also kills infected myocytes, and persisting viral RNA with disrupted dystrophin can leave a dilated ventricle even after the virus is gone.

· A positive parvovirus B19 result does not establish cause, because parvovirus B19 and HHV-6 genomes are among the most frequent viral findings in heart biopsies and are also found in hearts without myocarditis.

· Parvovirus B19 binds P antigen (globoside) on erythroid precursors and endothelium, and it injures small vessels.

Distinctions that matter

· Direct viral injury and immune-mediated injury are two phases of the same illness, days against weeks.

· Finding a viral genome in tissue is not the same as showing that it caused the disease.

Common misconceptions

· Reading troponin release as proof of coronary occlusion. Any process that kills myocytes releases troponin.

· Treating the T-cell response as purely protective. It clears virus and also kills infected myocytes.

Where it is taught: Section C06, paragraph on myocyte loss and the T-cell phase, the table of other myocarditis viruses, and the closing paragraph of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C06-Q02

Board-style question: The torn sarcolemma in coxsackie myocarditis

Correct answer: A. Viral protease cleavage of dystrophin, which links sarcomeric actin to the sarcolemma

Why it is right: Coxsackie B protease 2A cleaves dystrophin, the protein that links sarcomeric actin to the dystrophin-glycoprotein complex in the sarcolemma, and without that link the membrane tears under contractile stress, which is the same structural failure seen in Duchenne muscular dystrophy.

B is wrong: ADP-ribosylation of elongation factor 2 is the mechanism of diphtheria toxin, which arrests protein synthesis in an organism that stays in the pharynx rather than being recovered from stool.

C is wrong: Actin tails that push an organism from cell to cell are how Rickettsia rickettsii spreads through endothelium, and it is a tick-borne bacterium rather than a naked RNA virus.

D is wrong: Nonneutralizing antibody carrying virus in through Fc gamma receptors is antibody-dependent enhancement in dengue, which produces plasma leak rather than a torn sarcolemma.

A competent response contains

· A naked positive-sense single-stranded RNA virus from stool after a summer diarrheal illness, with myopericarditis, is coxsackie B virus.

· Viral protease 2A cleaves dystrophin, which links sarcomeric actin to the dystrophin-glycoprotein complex in the sarcolemma, so the membrane tears under contractile stress.

· The same protease cleaves eIF4G and shuts off cap-dependent host translation, which is a separate effect from the membrane failure.

Distinctions that matter

· Protease 2A has two targets, and only the dystrophin cut explains a torn membrane.

Common misconceptions

· Choosing the host translation shutdown as the cause of the tear. Stopping protein synthesis kills the cell over time but does not explain mechanical failure of the sarcolemma.

Where it is taught: Section C06 of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C07-BD1

Brain dump 1 · foundational retrieval: Diphtheria toxin and the tuberculous pericardium

A competent response contains

· Corynebacterium diphtheriae is a club-shaped gram-positive rod that lies in V and L patterns, contains metachromatic polyphosphate granules, and forms black colonies on tellurite agar; it colonizes the pharynx without invading deeply.

· Only strains carrying a lysogenic beta-prophage with the tox gene make toxin, and the bacterial repressor DtxR blocks tox transcription when iron is plentiful, so toxin is made when iron is scarce.

· Toxin mechanism: the B portion binds the heparin-binding EGF-like growth factor precursor, the toxin enters an endosome, acidification moves the A fragment into the cytoplasm, and the A fragment ADP-ribosylates elongation factor 2, so protein synthesis stops and the cell dies.

· Locally, dead epithelium, fibrin, and leukocytes form a gray adherent pseudomembrane.

· Absorbed toxin reaches the myocardium and peripheral nerves; myocarditis usually appears 1 to 2 weeks into the illness, often as the throat improves, with myocyte degeneration and necrosis and relatively little inflammation, causing atrioventricular and bundle branch block, arrhythmias, and heart failure, and it is a major cause of death in diphtheria. Demyelinating neuropathy follows later.

· Mycobacterium tuberculosis is an acid-fast rod whose mycolic acid-rich wall resists decolorization and lets it survive inside macrophages.

· It usually reaches the pericardium by retrograde lymphatic spread from mediastinal peritracheal and peribronchial lymph nodes, or by blood.

· The cell-mediated response forms caseating granulomas with a fibrinous, often bloody effusion, and as the inflammation heals the visceral and parietal layers fuse into a thick fibrous, sometimes calcified shell.

Distinctions that matter

· The toxin is an A-B exotoxin: the B portion decides which cells are reached and the A fragment does the damage.

· Diphtheria damages the heart without the bacterium leaving the throat, which separates it from every other organism in this chapter.

Common misconceptions

· Expecting plentiful iron to increase toxin production. DtxR blocks tox transcription when iron is abundant, so toxin is made when iron is scarce.

· Expecting heavy inflammation in diphtheritic myocarditis. There is myocyte degeneration and necrosis with relatively little inflammation, because the toxin kills the cell directly.

Where it is taught: Section C07, diphtheria and tuberculous pericarditis subsections of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C07-BD2

Brain dump 2 · integration and application: Predict the physiology of a fixed pericardium and a starved aortic media

A competent response contains

· Early diastolic filling is rapid until the ventricles meet the rigid shell, then stops abruptly, which produces a pericardial knock.

· The jugular venous pulse shows a steep y descent, matching that rapid early filling.

· The inspiratory fall in intrathoracic pressure is not transmitted through the shell, so the extra venous return of inspiration cannot enter the right heart and jugular venous pressure rises on inspiration, which is the Kussmaul sign.

· Diastolic pressures equalize in all four chambers because the shell fixes the total volume of the heart.

· Systemic venous congestion gives hepatomegaly, ascites, and edema.

· In syphilis the treponemes persist in the adventitia around the vasa vasorum, the small adventitial arteries that nourish the outer media of the thoracic aorta, and obliterative endarteritis narrows them, so the outer media becomes ischemic and loses elastic fibers and smooth muscle with patchy scarring.

· The ascending aorta and arch fail because they depend most on vasa vasorum; the intima over the scarred media wrinkles into a tree-bark pattern and may calcify.

· A dilated root and arch stretch the valve ring and cause aortic regurgitation, narrow the coronary ostia and cause angina, and a large aneurysm can compress the trachea, esophagus, or left recurrent laryngeal nerve.

· Atherosclerotic aneurysms favor the infrarenal abdominal aorta because atherosclerotic intimal thickening impairs diffusion from the lumen, which matters most where there are few vasa vasorum.

Distinctions that matter

· Constrictive physiology comes from a fixed total volume, which is why all four chambers equalize and why inspiration raises rather than lowers jugular pressure.

· The two aneurysm sites follow two different nutrient routes to the media: vasa vasorum in the thoracic aorta and diffusion from the lumen in the infrarenal aorta.

Common misconceptions

· Expecting jugular venous pressure to fall on inspiration as it normally does. The shell prevents the thoracic pressure change from reaching the right heart.

· Saying the treponeme destroys the aortic wall directly. The wall fails from ischemia of its media.

Where it is taught: Section C07, paragraph on constrictive physiology, the syphilitic aortitis subsection and flow, and the stop and predict box of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C07-RT01

Retrieval exercise: Rebuild the path from vasa vasorum to aneurysm

A competent response contains

· Step 1: treponemes persist in the adventitia around the vasa vasorum.

· Step 2: lymphocyte and plasma cell cuffs surround those vessels.

· Step 3: intimal thickening, which is obliterative endarteritis, narrows the vasa vasorum.

· Step 4: the outer media becomes ischemic.

· Step 5: elastic fibers and smooth muscle are lost, with patchy scarring.

· Step 6: the weakened ascending aorta and arch dilate.

· Step 7: aneurysm, aortic regurgitation, and coronary ostial narrowing follow.

· Obliterative endarteritis is concentric thickening of the intima of small arteries that narrows or closes the lumen and makes the tissue they supply ischemic; the cuffs around the vessels are lymphocytes and plasma cells.

· The media loses elastic fibers and smooth muscle, so the wall can no longer recoil against systolic pressure and stretches instead.

· On the cross section, the adventitia carries the vasa vasorum, the outer media is the layer that becomes ischemic, and the intima lies against the lumen.

· The loss of blood supply to the wall does the damage, not destruction of tissue by the organism itself.

Distinctions that matter

· The lesion is in the small adventitial vessels, while the failure is in the media they feed.

· The intimal tree-bark change is a consequence of the scarred media underneath it, not a primary intimal disease.

Common misconceptions

· Putting the endarteritis in the aortic intima. It is in the intima of the vasa vasorum.

· Expecting the aneurysm where the aorta is widest rather than where it depends most on vasa vasorum.

Where it is taught: Section C07, syphilitic aortitis flow and the paragraph after it of the CV-17 Cardiovascular Pathogens notes.

NPLEX-P1-CARD-17-01-C07-CA01

Clinical application: Chest discomfort and a widened ascending aorta in a man with a remote untreated infection

A competent response contains

· Recognize: Treponema pallidum in tertiary syphilis, usually 10 to 30 years after untreated infection, and the tree-bark intima overlies a media scarred by obliterative endarteritis of the vasa vasorum.

· Explain: the treponemes persist in the adventitia around the vasa vasorum of the thoracic aorta, lymphocyte and plasma cell cuffs form, intimal thickening narrows those vessels, the outer media becomes ischemic, and elastic fibers and smooth muscle are lost with patchy scarring; the leaflets are normal because the leak comes from dilation of the root stretching the valve ring rather than from disease of the leaflets.

· Differentiate: this aneurysm is in the ascending aorta and arch and arises from ischemia of the media through narrowed vasa vasorum, while atherosclerotic aneurysms favor the infrarenal abdominal aorta and arise from intimal thickening that impairs diffusion from the lumen; the feature that explains the difference is the density of vasa vasorum, which is richest in the thoracic aorta and sparse in the infrarenal aorta.

· Investigate: the organism is a thin spirochete seen by dark-field microscopy or silver stain and is invisible on Gram stain because it is too thin to resolve; in the wall, look for loss of elastic fibers and smooth muscle with patchy medial scarring, and in the adventitial vessels look for intimal thickening with lymphocyte and plasma cell cuffs.

· Decide: the exertional chest discomfort comes from narrowing of the coronary ostia by the dilated aortic root, and a large aneurysm of the arch can also compress the trachea, the esophagus, or the left recurrent laryngeal nerve.

· Justify: the outer media of the thoracic aorta is nourished by the vasa vasorum rather than by diffusion from the lumen, so narrowing those vessels starves the media of oxygen; the organism is scarce in the wall and does not destroy it directly, and the structural failure follows the ischemic loss of elastic fibers and smooth muscle, which is why the wall dilates under normal systolic pressure.

Distinctions that matter

· Aortic regurgitation here is a problem of the root rather than the leaflets, which is why the leaflets look normal at surgery.

· The wide pulse pressure and collapsing pulse follow from regurgitant flow back into the ventricle in diastole.

Common misconceptions

· Looking for the organism in the media. It persists in the adventitia around the vasa vasorum.

· Expecting Gram stain to show a spirochete. Treponema pallidum is too thin for it, so dark-field or silver stain is used.

Where it is taught: Section C07, syphilitic aortitis subsection, the flow, and the paragraph on consequences of the CV-17 Cardiovascular Pathogens notes.